

Why does therapeutic drug development take so long? Antibodies as drugs: Herceptin (part 3)

By Jeremy M. Berg PhD, Stand Up for Science Advisor

SEP 11, 2026

Last week I highlighted Gleevec, one of the first medicines specifically targeted to known genetic changes in certain types of cancer. Gleevec was more or less universally successful in chronic myelogenous leukemia in large part because all of the cancers of this type have essentially the same genetic change. In contrast, most other cancer types are quite heterogeneous with different subtypes of cancer in different individuals. Remarkably, one aspect of this heterogeneity provided a key insight that led pioneering oncologist Dennis Slamon to develop a treatment targeted to a subset of highly aggressive breast cancers. Moreover, this treatment was based on monoclonal antibodies, materials that were widely used as research reagents, but not previously used as drugs.

Breast cancer is a heterogeneous disease, at the genetic, cellular, and treatment levels. Most breast cancers, if found early when they are small, can be successfully cured with surgery and radiation treatment, sometimes in conjunction with drugs that help prevent recurrence. For most breast cancers, a major driver of tumor growth is the hormone estrogen. Estrogen binds to receptors within the cells that play central roles in gene regulation. Approximately 75% percent of breast cancers express the estrogen receptor, often along with the related progesterone receptor, at relatively high levels. These high levels of estrogen receptor expression make these cells very responsive to the growth-stimulating effects of estrogen. However, some breast tumors are not dependent on estrogen but, instead, rely on other pathways to support their rapid growth.

One such pathway involves a protein called HER2, an oncogene product first identified in 1984. HER2 protein is a member of the epidermal growth factor receptor family. These proteins are present in cell membranes with a domain on the outside of cells that can respond to molecules in the environment and also a protein kinase domain on the inside of the cells that can be activated based on extracellular signals. In 1987, oncologist Dennis Slamon discovered that some relatively intractable breast cancers express higher than normal amounts of HER2 on their cell surfaces.

Antibodies represent a unique way of targeting such proteins. Antibodies are proteins generated by the immune system to block infectious agents such as viruses and bacteria. Normally, the antibodies generated in response to an infection are heterogeneous mixtures. However, in 1975, scientists had developed methods for generating so-called "monoclonal antibodies" in mice. These are pure proteins, generated initially from a single immune cell. Monoclonal antibodies are powerful tools, used extensively in research. Slamon collaborated with scientists at Genentech to develop monoclonal antibodies that specifically bound HER2. To use these as potential drugs, these antibodies were "humanized", made to be more similar to human than mouse, without affecting their targeting, so that the antibodies themselves would be less likely to trigger an additional immune response.

In 1992, they first tested one of these antibodies, referred to as trastuzumab or Herceptin, in patients with metastatic breast cancer that had progressed despite other chemotherapy. Together with other chemotherapy, these antibodies were infused into the patients on a weekly basis and disease progression was monitored. Remarkably, some of the patients achieved decreased disease progression, longer survival, and even complete remission. even when given without additional chemotherapy.

After further trials, Herceptin was approved for treatment of HER2-positive breast cancer in 1998. The use of Herceptin has been refined over the decades since its development, both for primary HER2-positive breast cancer and for metastatic HER2-positive disease. It can be given either as a single agent or in combination with other chemotherapy. Remarkably, some of the first patients with metastatic disease treated with Herceptin have lived disease-free for decades after participating in the clinical trials. The success of Herceptin laid the foundation for the now widespread use of monoclonal antibodies as drugs.

